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Internationally mobile scientists as knowledge transmitters: A lexical-based approach to detect knowledge transfer

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Abstract

This paper explores the knowledge transfer of internationally mobile scientists. It builds upon previous work on the development of methods for detecting the knowledge transfer of German scientists. Using abstract terms of publications covered in Scopus, this paper proposes a lexical-based approach to identify knowledge transmitters. These scientists are characterized by acquiring knowledge from their co-workers during their international stay and transferring it upon return to German co-workers. Knowledge is operationalized as the co-occurrence of rarely used abstract terms. Knowledge transfer is expressed as the diffusion of these term combinations in co-authorship networks. The method developed was validated by contacting the bibliometrically identified knowledge transmitters and asking them what they believe they learned during their stay abroad. A control group of internationally mobile scientists without traceable knowledge transfer was similarly asked to report on their knowledge acquisition. The findings suggest that bibliometric data are capable of detecting knowledge transmitters among German scientists who were internationally mobile. The juxtaposition of the responses on their perceived knowledge acquisition and the bibliometrically identified lexical terms shows that the method proposed is well suited to studying the knowledge transfer of internationally mobile scientists. The strength of the method is its simplicity and high precision.

1 | INTRODUCTION

International mobility of scientists is associated with a plethora of positive effects on the careers of scientists, such as the expansion of international networks, the increase in scientific productivity and impact, the access to research infrastructures and funds, and the acquisition of scientific knowledge (Netz et al., 2020). Whatever the type of research mobility (international or national), it paves the way for collaboration and enables access to scientific networks, research infrastructures, cutting-edge technology, and place-specific phenomena of study (Ivancheva & Gourova, 2011; Williams & Baláž, 2008). In some disciplines, international mobility is considered a given academic convention (Bauder, 2012) or as an "expectation" in the academic labor market and part of the "excellence" requirement (Morano-Foadi, 2005). To rule out that mobility is rated higher than the quality of

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the mobility experience (Ackers, 2008), it is important to study the scientific benefits that international research experience can bring.

Even though knowledge transfer is regarded as a major benefit of working abroad, it has not yet been studied comprehensively enough (Netz et al., 2020). We still know little about how scientists engage in knowledge transfer activities and whether and how knowledge transfer can be tracked. Given the tacit and complex nature of knowledge, its actual transfer as well as the tracing of it is a challenge. Most previous studies that capture knowledge transfer use qualitative data and subjective measures gained from self-reported data. There are a few exceptions such as Gibson and McKenzie (2014), who used the number of conferences and seminars attended abroad as a proxy for knowledge transfer, or Horta et al. (2018), who quantify the multidisciplinarity of knowledge by counting the number of fields in which scientists publish. Seminal studies on knowledge transfer in the sociology of sciences can be traced to Collins (1974, 2001), who illustrated the crucial role international mobility plays in the transfer of tacit knowledge. Latour (1987) demonstrated that knowledge cannot be simply transferred without being changed and therefore introduced the concept of "translation."

Knowledge transfer between two scientists is an elementary process of knowledge flows in scientific communities. Since knowledge transfer cannot be directly observed, early bibliometric studies relied on citation linkages as a proxy for knowledge flows among scientists. Citation linkages between publications imply a flow of knowledge from the authors of the article being cited to the authors citing an article (Hassan & Haddawy, 2013; van Leeuwen & Tijssen, 2000; Zhuge, 2006). Citations are explicit links to earlier documents and constitute a paper trail that provides empirical evidence of the direction of knowledge transfer (Van Leeuwen & Tijssen, 2000). Due to the many motivations and reasons to cite a paper (Bornmann & Daniel, 2008; MacRoberts & MacRoberts, 1989; Moravcsik & Murugesan, 1975), citations can indicate knowledge transfer only to a certain extent. Moreover, citation linkages are not capable of indicating the specific knowledge transferred. The goal of this proof-of-principle study is thus to renounce the use of citation linkages as a proxy for knowledge transfer and to present a more advanced lexical-based approach to detect knowledge transfer of internationally mobile scientists.

A cornerstone of studying knowledge transfer of internationally mobile scientists is the identification of scientists who were working abroad using bibliometric techniques. The use of publication data to trace the mobility of scientists was introduced by Laudel (2003). Numerous studies that relied on WoS and Scopus data to track scientists' international mobility followed (Aman, 2018b; Conchi & Michels, 2014; Halevi et al., 2016; Moed et al., 2013). A comprehensive taxonomy for the study of international scientific mobility using bibliometric data was provided by Robinson-Garcia et al. (2019).

This paper is a follow-up study on the exploration of bibliometric methods to detect knowledge transfer inherent to international mobility. In a first step, I examined the reliability of the Scopus author ID to identify authors and tested whether it suffices to track international mobility of German scientists (Aman, 2018b). The comparison of mobility data from Scopus publications with CV-based data on residence countries showed that bibliometric data is suitable to identify a scientist's international mobility. Guided by the idea that whenever internationally mobile scientists interact with scientists abroad they become similar to one another in their choice of references cited and lexical terms used in abstracts, I approximated knowledge transfer using cosine similarity (Aman, 2018a). On the basis of abstract terms and references, the cosine similarity of internationally mobile authors and their co-authors was calculated, indicating in how far their research content becomes similar over time. While the similarity study did not consider the knowledge types transferred, a following proof-of-concept study tested two different approaches (Aman, 2020). Published knowledge was operationalized as the use of rarely cited publications, whereas methodological know-how was operationalized as the occurrence of lexical terms in abstracts of publications (Aman, 2020).

Based on the exploratory results of these previous approaches, this study traces specific scientific knowledge by operationalizing it as the occurrence of rare lexical terms in abstracts of publications. The operationalization of knowledge transfer relies on a network-based approach to identify knowledge transmitters and is thus similar to a previous study (Aman, 2020). The method presented enables the identification of German scientists who act as knowledge transmitters, in the sense that they acquire knowledge during their international stay abroad and transfer that knowledge to their home country upon return. The findings gained from the method developed were validated by contacting scientists with and without a traceable knowledge transfer and asking them what they believe they learned abroad. Their replies on scientific activities and perceived knowledge acquisition were juxtaposed with the bibliometrically identified lexical terms representing the transferred knowledge. The validation yields promising results and valuable insights into the knowledge acquired and transferred by internationally mobile scientists.

The following section provides a theoretical background discussing the importance of international

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mobility and scientific collaboration in knowledge transfer processes.

2 | THEORETICAL BACKGROUND

Scientists are bearers of experiences, skills and knowledge, which they exchange consciously or unconsciously in everyday work. Some types of knowledge exist only in specific places and at certain times and therefore require the co-presence of scientists in order to be transferred. The attempt to transfer knowledge from source to target is usually driven by the ambition to recreate observed research practices or results. Whenever scientists transfer knowledge between source and target, they shape and modify that knowledge. Thus, the transfer of knowledge involves transformation, which implies that it is impossible to simply replicate knowledge processes (Latour & Woolgar, 1986). Moreover, the transfer process leads to new and unique local versions of knowledge (Latour & Woolgar, 1986). The alteration of knowledge in transfer processes results from the richness of interpretations that knowledge can trigger in each actor within a network (Latour, 1987). Thus, knowledge transmitters and recipients of knowledge are not passive actors, but active translators of knowledge (Latour, 1987). However, the extent to which knowledge is adopted and integrated in the recipient's research practice may vary and also depend on the knowledge type.

A basic distinction of knowledge types goes back to Polanyi (1966), who distinguished tacit and explicit knowledge. Whereas tacit knowledge is intuitive, often unarticulated and hard to verbalize, explicit knowledge can be articulated in formal language and easily transferred among individuals. However, knowledge is not purely explicit or tacit but rather exists along a continuum ranging from tacit to explicit knowledge (Nonaka & Von Krogh, 2009).

A suitable distinction of three scientific knowledge types in regard to their access was provided by Gläser (2006, pp. 108–113). *Published knowledge* is easily available to the scientific community and can be transferred in the absence of physical presence. *Informally communicated knowledge* can be communicated on demand between scientists whenever they interact in co-presence or virtually, using information and communication technologies. In contrast, *tacit knowledge* is highly individualized and can hardly be expressed verbally. Depending on the scientific field, tacit knowledge requires face-to-face interaction and observation of epistemic practices. Previous studies show that geographic mobility is crucial for the transfer of tacit knowledge because it enables personal contact, observation of colleagues and interaction (Collins, 1974; Collins, 2001; Jöns, 2009). International mobility may become a necessity if the knowledge to be gained is tied to individuals, research environments or large-scale facilities located abroad. One major function of international mobility is to gain access to research groups that are crucial units in the production of knowledge (Gläser & Laudel, 2001). Scientists may wish to go abroad in order to collaborate with scientists whom they admire or wish to work with scientists who bear unique skills and knowledge.

Geographic mobility thus plays an important role in the acquisition and recombination of knowledge (Gläser, 2006; Laudel, 2003). Internationally mobile scientists bring their knowledge to a host research group and affect the discussion and combination of elements of existing knowledge. After a successful acquisition and application of new knowledge at the host research group, internationally mobile scientists can bring that knowledge back to their home country and make it accessible to new co-workers.

An internationally mobile scientist can act both as a sender and as a recipient of knowledge while working abroad. As in previous work (Aman, 2020), the present study focuses on internationally mobile scientists as recipients of knowledge while being abroad and as senders of knowledge upon return in their home country. Thus, internationally mobile scientists acted as knowledge transmitters between a source of knowledge abroad and a recipient of knowledge in their home country.

Figure 1 provides a general model of knowledge transfer that is not necessarily tied to international mobility. S represents the source of knowledge, T the transmitter and R the recipient of knowledge. It is assumed that S generates or acquires knowledge in the pre-transmission phase, before an interaction takes place between S and T. During the transmission phase, S and T interact so that knowledge can be transferred from S to T. In the post-transmission phase the interaction of T and R enables the transfer of knowledge from T to R. Thus, there are two consecutive processes of knowledge transfer (S to T and T to R) and T acts as a transmitter of knowledge from S to R.

Taking international mobility into account, the transmission phase equals the mobility phase, whereby the internationally mobile scientist acts as a transmitter of knowledge between the source abroad and a recipient in the home country.

As the focus of this study is on tacit knowledge, its transfer can be best described with the concept of "knowledge conversion" that refers to the iterative alternation of knowledge between the tacit and explicit dimension (Nonaka & Takeuchi, 1995). The alternation of these knowledge types is described in the SECI model

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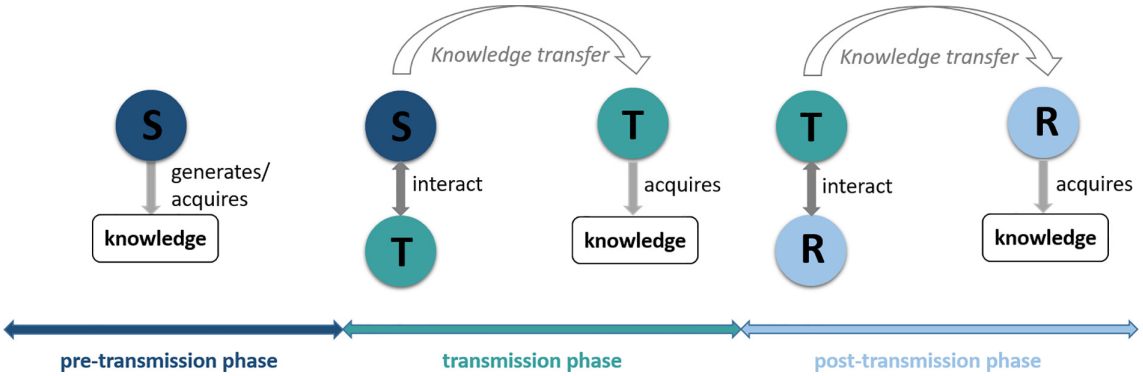

FIGURE 1 Theoretical model of knowledge transfer from source of knowledge S to transmitter T to recipient R

encompassing the four processes of socialization, externalization, combination, and internalization. Even if the SECI model by Nonaka and Takeuchi (1995) results from research in knowledge management and not the sociology of science, it can be adapted to specify the transfer of tacit knowledge related to international mobility. The socialization step in the underlying model refers to the joint research work abroad that enables observation and interaction and thus the transfer of tacit knowledge. The externalization step relates to the conversion of tacit knowledge practiced in the research group into explicit knowledge that is informally communicated or covered in publications. The combination step refers to the creation of knowledge via informal communication in discussions among research group members. A recipient of the knowledge such as the internationally mobile scientist can adapt and modify the transferred knowledge. In the last stage of internalization, the explicit knowledge is converted by the internationally mobile scientist into a tacit form. This internalization process is related to "learning by doing" and requires thus co-presence which can be achieved by international mobility.

The expectation, the need and the value of mobility differ between disciplines. Whereas some disciplines involve locally contextualized knowledge and tend to be "place-specific," other disciplines make universal knowledge claims and rely on high standardization (Jöns, 2007). Thus, disciplines in which knowledge has a low level of standardization and is bound to research infrastructures, local phenomena or even individuals, mobility is a more important factor for the transfer of knowledge. Knowledge is being increasingly created in research groups consisting of members with different scientific and professional background. Their capacity to innovate and deal with complex tasks exceeds that of individuals (Hautala, 2011; Nonaka & Takeuchi, 1995). Joint research publications resulting from successful integration of internationally mobile scientists into research groups abroad are likely to be indicators of knowledge transfer (Laudel, 2002). The co-authorship of internationally mobile scientists with scientists from their host research group can facilitate the transfer of *published knowledge* by hinting at publications, the transfer of *informally communicated knowledge* by discussing best practices of a research approach, and the transfer of *tacit knowledge* by indicating that it exists at a certain particular place (Gläser, 2006). As co-authorship implies a preceding successful collaborative research process with opportunities to exchange different types of knowledge, the method to be presented relies on co-authorship networks.

3 | DATA AND METHODS

3.1 | Database to identify internationally mobile German scientists

The identification of scientists here relies on the Scopus author ID (Elsevier), a powerful algorithm that aims to combine all publications of an author under a single ID to handle common first and last names. Previous studies report that the Scopus author ID enables reliable author name disambiguation (Aman, 2018b; Conchi & Michels, 2014; Moed et al., 2013). International mobility is defined as a research stay at a host institution outside the scientist's home country. Internationally mobile scientists are those whose affiliation in publications changes from one country to another. The study considers international stays of up to 3 years because longer stays are associated with greater opportunity to interact with scientists enabling the transfer of more complex and tacit forms of knowledge (Edler et al., 2011).

As the scientists in the underlying dataset have mainly published with German affiliations between 2007 and 2015, they are referred to from here on as *German scientists*, irrespective of their nationality. I distinguish a pre-mobility phase ranging from 2007 to 2009, in which

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scientists published exclusively from German institutions; a mobility phase between 2010 and 2012, in which they were abroad, and a post-mobility phase ranging from 2013 to 2015, in which they were again exclusively affiliated to German institutions. The choice of the three time periods, the demand for continuous publication over three periods, and a mobility phase reduces the size of the sample to 476 scientists.

3.2 | Database to identify knowledge transmitters

To capture the knowledge transfer of internationally mobile scientists, I focus on abstracts of their publications and those of their co-authors in 2007–2015 as covered in Scopus. The Scopus data is integrated in a licensed in-house database held by the *Competence Centre for Bibliometrics*[1], facilitating extensive computations.

The set of abstracts comprises more than 2 million Scopus publications and is referred to as the *corpus*. In a first step, all abstract terms in the corpus were automatically extracted. The terms extracted were lowercase and consisted only of alphabetical letters. The publication id, the term and the number of occurrences of a term per publication were determined. In a following step, the term frequency (TF) and the inverse document frequency (IDF) of the abstract terms were computed on the basis of the corpus. The product of TF and IDF determines the weight of a term. The higher the score the rarer the term and vice versa.

To operationalize knowledge transfer, I restricted the corpus to abstract terms with a relatively high score. The number of distinct terms in the corpus was 68,242 and the TF-IDF score ranged from 0.01 (e.g., *of* or *the*) to 14.65 (e.g., *chloropropionates*, *dictyophycus*, or *zontivity*). To filter out stop words such as *the*, *from*, or *with* that bear no knowledge, I restricted the terms to a score higher than 5.0. Examples of terms with a score of 5.0 in the corpus are: *fourier*, *lymph*, or *semiconductor*. Restricting the overall number of terms in the corpus (852,074) to those with a TF-IDF score greater than 5.0 reduced the dataset to 367,816 terms. Thus, only 43.18% (367,816/852,074) of all terms in the corpus were considered suitable to indicate specific knowledge due to their rarity.

3.3 | Operationalization of knowledge and knowledge transfer

The specific knowledge whose transfer I want to trace is operationalized as the co-occurrence of two of these rare terms in a single abstract. To detect the transfer of knowledge, the first publication year of the co-occurrence of two terms was determined. The limitation to rarely used abstract terms suggests that lexical terms can be traced back to co-workers abroad who act as sources of knowledge and are not picked up individually by internationally mobile scientists. Co-workers are defined as co-authors of the mobile scientists at the same institution. The identification of co-workers of German scientists relies on the institutional coding that exists for all German institutions and is regularly updated (Winterhager et al., 2014).

In the following section, a definition of traceable knowledge transfer is provided. Two-term combinations, which were used by a co-worker abroad (source) between 2007 and 2009, thus, before an interaction had taken place between the mobile scientist and a co-worker abroad, are indicative of knowledge transfer. Moreover, a specific two-term combination must be used for the first time by the mobile scientist in publications affiliated with the host institution on the condition that they are not co-authored with the source of knowledge. Furthermore, it is required that the internationally mobile scientist has at least one co-authored publication with a co-worker in Germany in the post-mobility phase (2013–2015). If this German co-worker uses the same two-term combination for the first time in publications of their own, they are considered as a recipient of knowledge. Only the strict order of usage of term combinations allows the detection of knowledge transfer from source to transmitter to recipient.

Figure 2 operationalizes the theoretical model of knowledge transfer in Figure 1 by specifying the knowledge (two-term combination represented as Term a + Term b), the mode of knowledge transfer (co-authorship) and how it manifests (in publications) when internationally mobile scientists act as knowledge transmitters. The lighter the color the later the year in which the knowledge was transferred from one node to another.

Node S represents the source of knowledge and thus a future co-author of the internationally mobile scientist T. It is assumed that S uses a two-term combination between 2007 and 2009, thus before a detectable interaction between S and T. During mobility, S and T interact and co-author at least one publication that is not necessarily related to the two-term combination. While abroad, T uses the same two-term combination for the first time in a publication of their own. In the post-mobility phase, there has to be at least one co-authored publication of T and R that is not necessarily related to the two-term combination. Moreover, the German co-worker is considered as a knowledge recipient R only if they use the same two-

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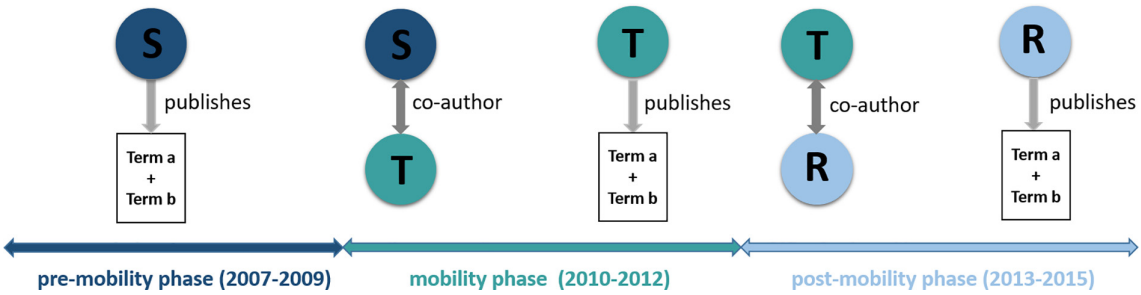

FIGURE 2 Illustration of knowledge transfer from source of knowledge S to transmitter T (internationally mobile scientist) to recipient R

TABLE 1 Overview of number of scientists with traceable and without traceable knowledge transfer

	Number of scientists identified via Scopus author id	Scientists with available CV and e-mail	Number of respondents	Response rate
Scientists with traceable knowledge transfer	111	54	28	51.9%
Scientists without traceable knowledge transfer	365	71	28	39.4%
Total	476	125	56	44.8%

term combination for the first time in a publication of their own. The internationally mobile scientist functions as a transmitter T between the source abroad and the recipient in Germany.

3.4 | Validation of bibliometric findings

The bibliometric method was primarily developed to identify internationally mobile scientists who function as knowledge transmitters between the country of mobility and Germany. However, the method also enables the identification of internationally mobile scientists who do not act as knowledge transmitters. The method was validated by contacting these two groups of scientists and asking them what they believe they learned during their stay abroad. Emails are a particularly useful instrument because they reach scientists in a quick and efficient way. They enable obtaining open answers that may comprise a few buzzwords up to sentences. In this study, emails worked well because they addressed only one specific question. Note that the emails were sent in German and all responses were in German, except for some English keywords. Due to restrictions in acquiring the appropriate data, the set of scientists who have been abroad and acted as knowledge transmitters is rather low. Table 1 provides an overview of the number of scientists identified with and without traceable knowledge transfer.

Even though 111 scientists were identified as knowledge transmitters, the number of knowledge transmitters with an email address available and a CV corroborating that the international mobility accorded with the bibliometric data is 54. These scientists were contacted and a reminder was sent a few weeks later. After contacting these scientists up to twice, 28 responses were obtained. In comparison, the set of internationally mobile scientists who did not act as knowledge transmitters is larger (365). Therefore, only as many non-transmitting scientists were contacted up to two times (71), until the same number of responses (28) was reached. Thus, the group of respondents with traceable knowledge transfer is as large as the group of scientists without a traceable knowledge transfer. To reach the same number of responses in these two groups, a personalized email had to be sent to 125 scientists in total. The response rate in the group of knowledge transmitters was 52%, and 39% in the group of scientists without a traceable knowledge transfer. The information was self-reported which may raise concerns in regard to response accuracy.

4 | RESULTS

4.1 | Descriptive analysis

The results section starts with a descriptive analysis of all 125 scientists in the study, and is followed by the validation of the bibliometric method.

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Figure 3 shows the total number of scientists by discipline. On the basis of their recent and past affiliations, scientists were assigned to the Fields of Science and Technology (FOS) classification (OECD, 2002). The majority of internationally mobile scientists work in the Natural Sciences, followed by Medicine, and Engineering and Technology.

Besides the specialty of the scientists, the length of their mobility was analyzed. In most cases the CVs of scientists provided information on the duration of their stays, especially in combination with a fellowship. The visits were divided into short-term stays of 2–6 months, medium-term stays of 7–12 months, long-term stays of 13–24 months, and research activities of a duration of up to 4 years. A distinction between short-term, medium-term and long-term stays is associated with different levels of effort to cope with the new research environment and culture. Short-term activities of up to 6 months are comparatively easy to realize as temporary accommodation is easy to find. Moreover, it may not require the application for a visa or the management of other bureaucratic processes.

With respect to the duration of stay, it is evident that in the group of respondents with traceable knowledge transfer, the majority of scientists was abroad between 13 and 24 months, whereas in the group of respondents without traceable knowledge transfer the majority was working abroad for 7–12 months. A particularly interesting finding is that the actual durations of the stays abroad were mostly shorter than the bibliometric data reflected. The bibliometric identification of mobility relies on publication years and can indicate mobility only on a yearly basis. Moreover, the juxtaposition of CV and bibliometric data shows that a research stay abroad of 1 year may result in articles that are spread over three publication years. This distortion can be caused by publication delays between submission and final publication or by the further use of the host affiliation, because the majority of the research work for a paper was conducted at the research institution abroad (Figure 4).

4.2 | Validation of respondents with traceable knowledge transfer

To validate how the bibliometrically identified knowledge transfer relates to scientists' perceptions about their knowledge acquisition, the two-term combinations were compared against their replies. Judging whether the term combinations match the email responses could be considered subjective, but it was a reasonable way to validate the bibliometric findings.

The results of the validation are illustrated by providing the reply of an internationally mobile scientist and the network graph with two-term combinations. In the first example, a computer scientist stayed as a postdoc at the Carnegie Mellon University (Pittsburgh, Pennsylvania) and replied briefly that he worked on "robotics, security certification and autonomous driving." The method identified overall four two-term combinations in abstracts representing knowledge transfer. Figure 5 juxtaposes the table with the two-term combinations and a network graph derived from it. Each node represents a term. The larger a node the more often the term occurs, independent of the other term in the combination. A node is connected to another node only if the terms represented by these nodes co-occur in an abstract. Hence, two-term combinations are always connected by an edge.

The size of the nodes "vehicle" and "road" reveals that these terms appear more often than other terms. This first example shows that the two-term combinations match the scientists' perception of knowledge acquisition

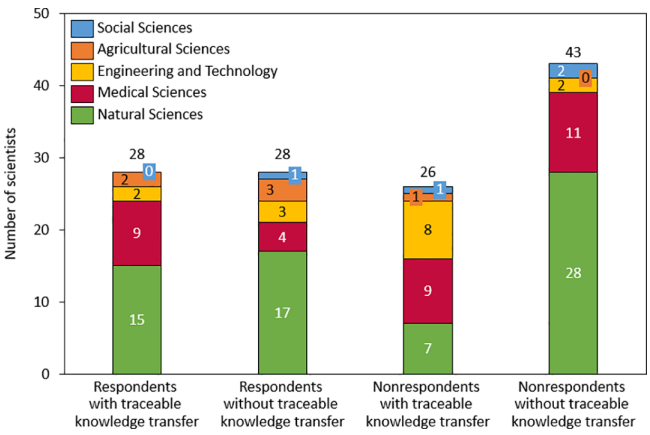

FIGURE 3 Respondents with and without traceable knowledge transfer and nonrespondents with and without traceable knowledge transfer according to OECD fields

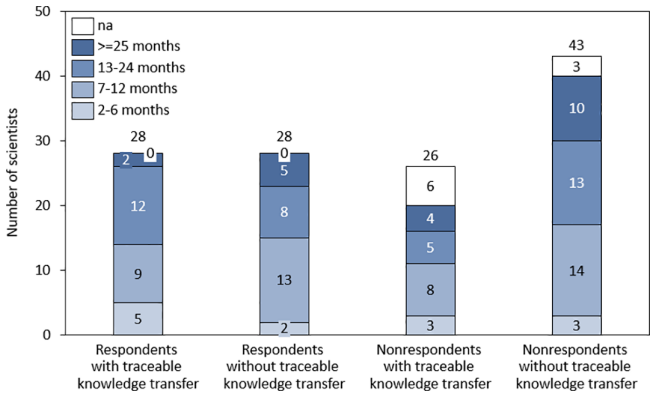

FIGURE 4 Duration of international stay of respondents with and without traceable knowledge transfer and nonrespondents with and without traceable knowledge transfer

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Table

Term a	Term b
autonomous	road
controller	disturbances
obstacles	vehicle
road	vehicle

Network graph

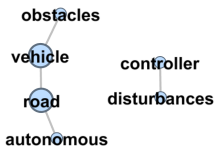

FIGURE 5 Table and network graph with two-term combinations indicating the knowledge transfer of a German computer scientist

abroad and reflect his short reply. Another scientist was staying at the Department of Chemistry in Berkeley (University of California) and replied:

> I am delighted to explain briefly what I believe to have learned during my stay. I would like to distinguish between technical and scientific skills:
> Technically, I learned on the one hand photoelectron and ion spectroscopy with imaging detectors (e.g., velocity map imaging VMI) and on the other hand the generation of so-called high harmonic generation (HHG) with intensive femtosecond lasers.
> But the most important thing for me was the "scientific" side. The environment in Berkeley gave me the opportunity to get to know many other scientific questions and experiments and to discuss them with colleagues.

Figure 6 shows the two-term combinations that were first used by the internationally mobile scientist while working abroad and upon return by future co-workers in Germany. There are several components from two to six connected nodes.

The specificity of the terms varies from general terms such as "indirect" to highly specific terms such as "Rydberg." The network graph represents well what the scientist described in his email. He explicitly mentioned the terms "photoelectrons" and "femtosecond" in femtosecond laser. However, the example also demonstrates a weakness of the method. The terms of the phrase "ab initio," describing a method in nuclear physics, are separated. Despite this limitation, the network graph can unveil knowledge transfer of which international scientists were either not aware or which would not have occurred to them.

Another example illustrates what a physicist who stayed as a postdoc at the Department of Physics and Astronomy remembers to have learned: "central key words for the knowledge acquired during my stay at the UPenn are jamming, glassy dynamics." To be able to

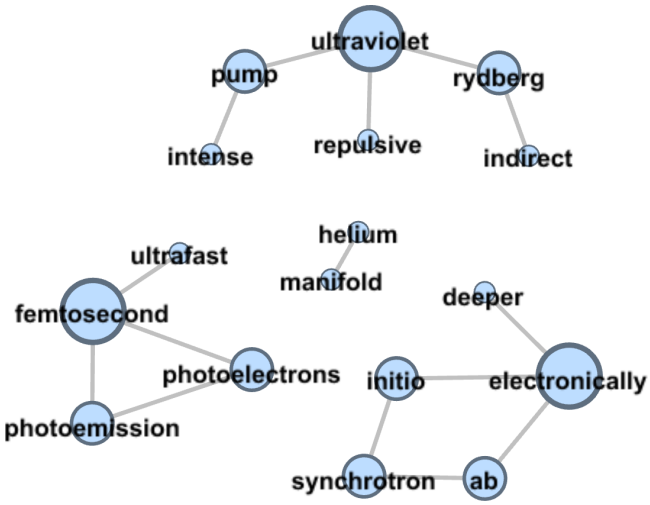

FIGURE 6 Network graph indicating the knowledge transfer of a German chemist

decide whether the terms reflect the knowledge acquisition, it is helpful to immerse oneself in the specialty. Glassy dynamics describes the dynamical processes in supercooled liquids and glasses. These processes are often encountered in systems with a large number of metastable configurations. With some background knowledge it is possible to interpret the knowledge transferred by the German physicist in Figure 7.

In another example, a physicist was working as a postdoc at the Lawrence Berkeley National Laboratory (LBNL):

> In a nutshell, what I learned at the LBNL: I gained a little insight into heterogeneous catalysis, especially on metal nanoparticles. Furthermore, I deepened my knowledge of scanning electron microscopy. I practiced independent work, also in the dynamic US research system .

The network graph of his two-term combinations is illustrated in Figure 8. There are several components that provide information about the co-occurrence of abstract terms. The terms in the network graph seem to match the terms the physicist used in his email.

The network graph shows that the scientist transferred knowledge primarily on STM (Scanning Tunneling Microscopy), photoelectrons and on QCM (Quartz Crystal Microbalance) with dissipation monitoring. However, the network graph also demonstrates the weaknesses of the method described above. The terms "owner" and "societies" do not provide information about the research content but the journal in which a paper was published. The original line at the end of the abstracts reads as

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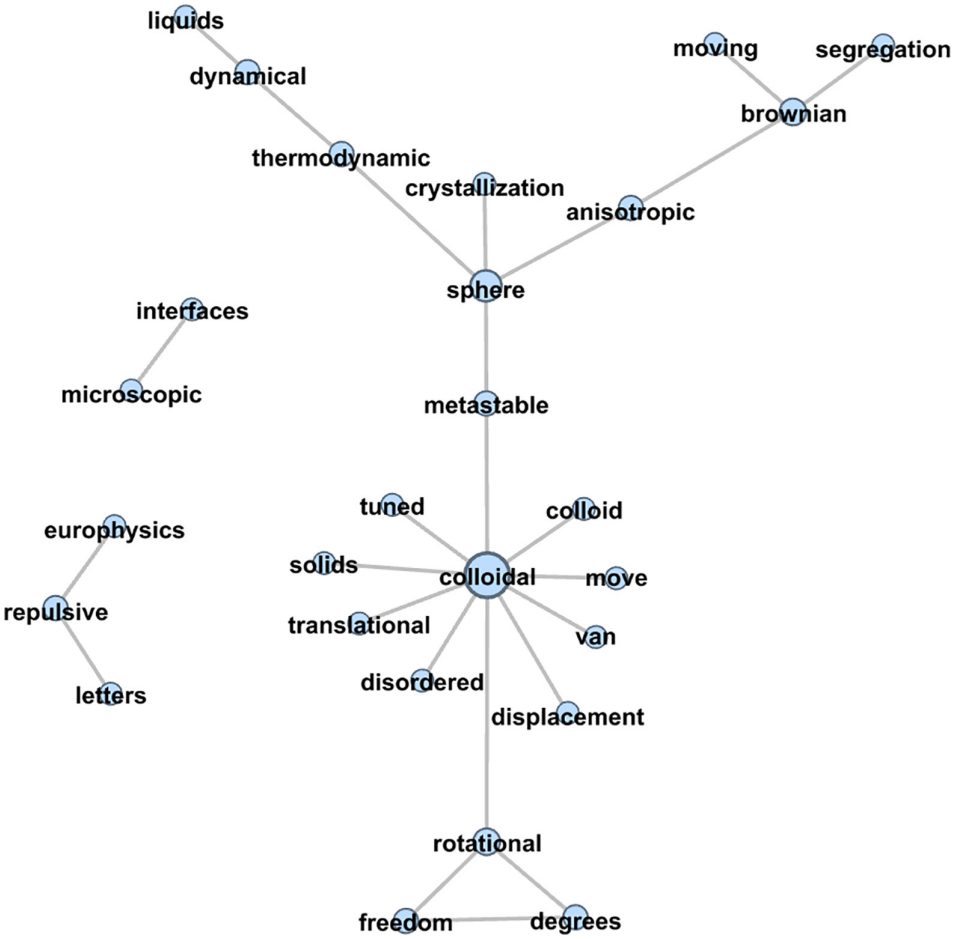

FIGURE 7 Network graph indicating the knowledge transfer of a German physicist

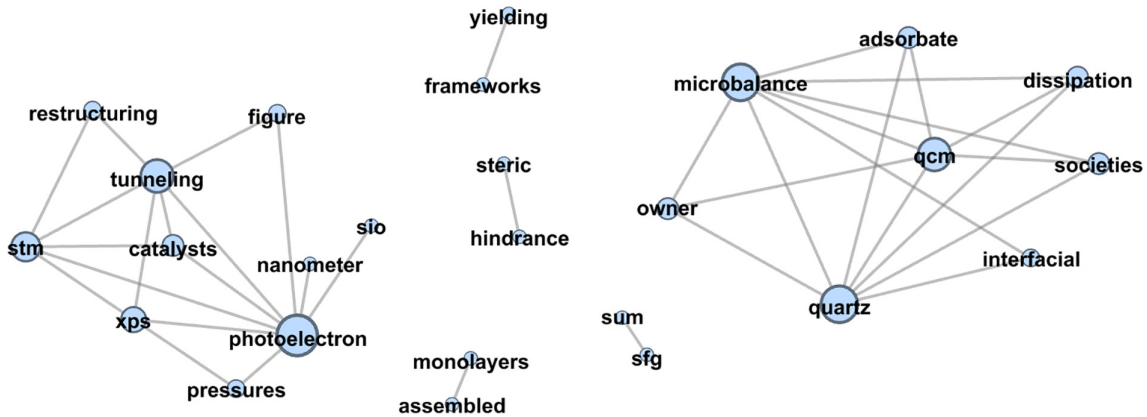

FIGURE 8 Network graph indicating the knowledge transfer of a German physicist

follows: "This journal is the Owner Societies [year]" and belongs to the journal *Physical Chemistry Chemical Physics*.

These examples serve to illustrate that the bibliometric method developed is imperfect but capable of indicating knowledge transfer at least to some degree. The simplicity of the method in combining two rare terms may come at the expense of precision. To quantify the precision of the two-term combinations, the combinations were manually verified for all 28 knowledge transmitters. The numbers of two-term combinations vary from 1 to 353 with an average of 43 combinations per knowledge transmitter. The combinations were coded with 2 if both terms were associated with the specialty of the scientist, with 1 if only one of the terms was associated with the specialty and with 0 if both terms were regarded as noise because they are too general. The values of a scientist were added and divided by the value count to

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determine the average rate that can range between 0 and 2. Moreover, the division of the average rate by two determines the precision of the two-term combinations. The precision of all scientists ranged between 89.1% and 100% with an overall average of 98.6%. Figure 9 illustrates the relationship between the number of two-term combinations per scientist and the precision of the term combinations. Each of the 28 bars represents a respondent with traceable knowledge transfer. The figures in the bars (x-axis left) indicate the number of two-term combinations per knowledge transmitter, whereas the precision is represented by the bars as percentages (right x-axis).

Figure 9 reveals a correlation between the number of two-term combinations and their precision. The fewer term combinations there are the more precise they are. The noise accumulates with an increasing number of term-combinations. One example of noisy two-term combinations in the dataset coded with 1 is "ab-synchrotron." Whereas "synchrotron" is a meaningful term, "ab" bears no information value without the term "initio" (cf. Figure 6). An example of a term combination coded with 0 is "owner-societies" (cf. Figure 8). Apart from the terms "owner" and "societies," the only other noisy term combinations in the dataset result from the three following terms: "ann," "mary," and "liebert." A look in the abstracts shows that the byline says: "Mary Ann Liebert, Inc. *is a* privately held independent publishing company." On the one hand, these terms are noise and more effort is required to exclude them with advanced natural language techniques. On the other hand, these seemingly erroneous terms also reveal that knowledge about the journal, in which study results are best suited, has been transferred.

Among the 28 respondents with traceable knowledge transfer, only one scientist replied that he has not learned anything abroad:

> Thank you for your interest in my stay in [city] 2009. It was a 6 month stay. During my stay I finished manuscripts and did a little teaching. I didn't learn any new methods or other things there. Yes, I learned a little Chinese. However, I analyzed the sample material that I had prepared there only back at my "home" University in Germany.

Even if the biologist denies to have acquired knowledge abroad, the bibliometric method reveals 91 two-term combinations. It is possible that the scientist moved to China in the middle of his research process and the two-term combinations rather provide information about novelty in his publications. Hence, a new research environment and new research objects may have caused the use of novel terms in abstracts that can be traced back to local co-workers. Thus, the bibliometric method may erroneously indicate knowledge transfer although a change in research work has little to do with learning from co-workers abroad. Another possible reason for the discrepancy between bibliometric data and email responses could be that scientists downplay their knowledge acquisition or are simply not aware of it.

4.3 | Validation of respondents without traceable knowledge transfer

Among the 28 respondents without traceable knowledge transfer, all but one scientist reported to have learned at least something abroad. One scientist disclosed in confidence that the research work associated with the foreign affiliation was mainly conducted at the German home university, but finished during the stay abroad. Another

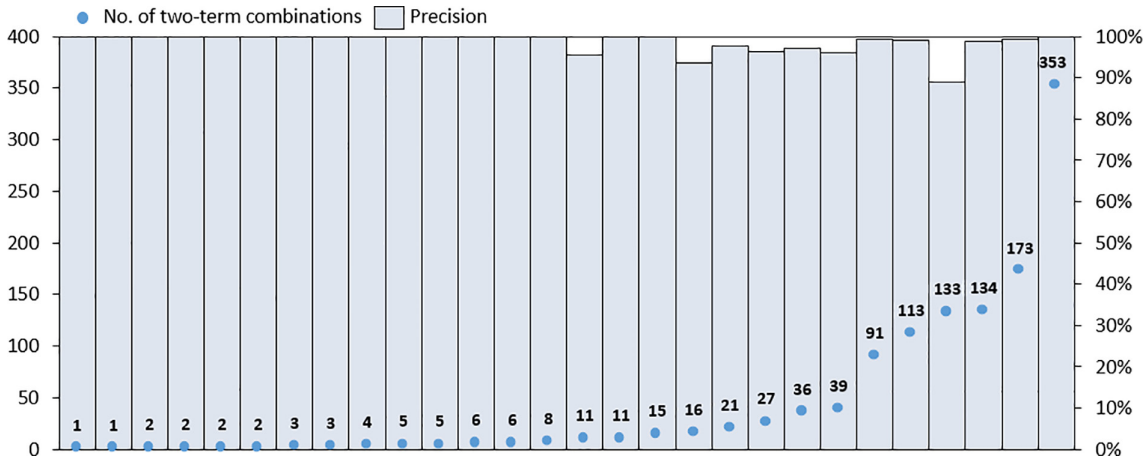

FIGURE 9 Overview of the relationship between the number of two-term combinations (left y-axis) and their precision in % (right y-axis) per knowledge transmitter (represented by a bar)

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scientist replied that it is more important to question what the research group abroad has learned from him and not what he learned during his stay. In the following section, it is discussed why the bibliometric method was not able to detect knowledge transfer, even though scientists claimed to have acquired knowledge abroad. One reason was that some scientists moved to the industry sector after their mobility experience. Thus, there was no opportunity for knowledge transfer to future co-workers in the academic sector. Another reason that prohibits the detection of knowledge transfer is that there are no German co-workers in the post-mobility phase who published on their own. Thus, potential knowledge recipients in the post-mobility phase co-authored each of their publications with the internationally mobile scientist. Another restriction that forbade the detection of knowledge transfer was the limitation of the post-mobility phase to 3 years. It is not excluded that knowledge is passed on after this 3-year period. A preventable reason for the erroneous non-detection of knowledge transfer resides in a too strict TF-IDF value. A scientist was working as a postdoc at the *Laboratoire de Physico-Chimie des Matériaux Luminescents in Lyon and replied*:

> I have used/learned Raman spectroscopy to examine the structure of glasses of different origins (natural: e.g., tektite or manufactured: e.g., by means of lasers).

Her abstracts show that she was publishing on Raman spectroscopy. Whereas the term "raman" has an IDF of 4.82, the term "spectroscopy" has a value of 3.29. The TF-IDF threshold was set to 5. Machine-based learning techniques may be better suited to avoid non-detection of knowledge transfer caused by a too strict TF-IDF threshold. A further reason for non-detection of knowledge transfer is the direction of knowledge transfer. A physicist reported that he rather transferred his knowledge to the research group at the *Institute Curie* in Paris:

> The decisive factor for me was the inclusion of a new topic (physics of cancer growth), in combination with bringing my methodology (minimalist mechanical simulations) into a group that had not used these methods before.

A final reason why no knowledge transfer was detected is that there was probably no interaction at the host research institution. A scientist was staying at the *Faculty of Health, Medicine and Life Sciences* at Maastricht University and briefly replied what she has learned:

> Submit an ERC application, work on my own theory, which I only published 3 years later.

In the final result section, the knowledge types acquired by the respondents with and without traceable knowledge transfer are presented.

4.4 | Knowledge types and acquisition of soft and social skills

Most respondents with traceable knowledge transfer (24) claimed to have either acquired methodological know-how or applied methodological knowledge. The replies of the remaining four scientists rather relate to the acquisition of soft and social skills than the acquisition of scientific knowledge. Overall, one in two scientists reported to have improved their soft and social skills. Most often, respondents mentioned that they met new people and expanded their network. Many scientists pointed out that these contacts are still of importance today. Apart from the collaboration and networking aspect, scientists replied that they enjoyed working on something completely new after 3–4 years of a doctorate. Positive features of their stay were to get out of their comfort zone and to be confronted with a new research landscape and a different research discussion. Important skills that were mentioned are the improvement of publication strategies, the writing of proposals, leadership skills and the accomplishment of independent work.

Most respondents without traceable knowledge transfer (23) also claimed to have either acquired methodological know-how or applied methodological knowledge. Two scientists reported to have acquired theoretical knowledge. Whereas one of the remaining scientists claimed not to have learned anything abroad, the replies of two scientists relate to the acquisition of soft and social skills. Overall, 10 respondents without traceable knowledge transfer mentioned that they benefited from their stay abroad in regard to the acquisition of soft and social skills. The responses are mostly similar to those of the knowledge transmitters.

5 | DISCUSSION

Despite the important role international mobility plays in the careers of scientists, knowledge transfer as a major positive outcome of international experience has been insufficiently studied. This is undoubtedly not due to limited interest but rather due to an absence of suitable methods to trace knowledge transfer. The purpose of this

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proof-of-principle study was to show that a lexical-based approach is able to detect knowledge transfer. In contrast to citation-based approaches the use of lexical terms is better suited to trace the transfer of tacit knowledge.

In the first instance, it was shown that it is possible to detect internationally mobile scientists with bibliometric data alone. Moreover, the applied method is capable of identifying knowledge transmitters among German scientists who adapt knowledge abroad and transfer that knowledge upon return to their co-workers in Germany.

To validate the bibliometric results, scientists with and without traceable knowledge transfer were asked to report what they have learned during their stay abroad. Hence, the two-term combinations were validated with email responses of internationally mobile scientists. Even though the data-set used in this study may be small, it provides valuable insights into knowledge transfer processes, considering that no explicit decision to engage in knowledge transfer was claimed by the scientists. It becomes apparent that the combination of automatically generated bibliometric data and textual data from emails are well suited to uncovering what German scientists learn during their stays abroad and transfer to future German co-workers.

It is notable that the dataset encompasses mostly natural scientists who worked in the United States. As the examples of the two physicists have shown, in some areas of physics, for example, particle physics, the co-presence with machines (large scale infrastructure) is necessary in order to conduct experimental research. Moreover, specific approaches, skills and knowledge that are bound to individuals require co-presence and thus geographic mobility to be transferred.

In regard to long-lasting effects of mobility, most scientists stressed that their international experience was utterly positive for their career and that they still benefit today. The positive effects are numerable and relate to aspects of scientific knowledge, communication skills and how to cope in a new research environment. While the detection of social and soft skills by bibliometric data alone is out of range, the method presented can detect scientific knowledge that is published in articles covered by Scopus.

The method presented relies on a TF-IDF-based approach, which is intuitive in the weighting of terms by determining the most important abstract terms and stop words that can be removed.

The calculation of TF-IDF could be improved by taking only publications from a particular discipline into consideration. Terms from specialties with a low coverage in Scopus may result in high TF-IDF values even though they are common in everyday language. In contrast, terms with high TF-IDF values may not necessarily represent the content of an abstract. However, a solution would require the proper delineation of specialties, which remains a problem.

The disadvantage of a TF-IDF-based approach is that it does not capture the position of terms in an abstract and is not able to capture semantics. Machine-based learning approaches for Natural Language Processing could be a good solution to derive the context of abstract terms and to provide semantic meaning. ML-based approaches would account for word order and noun phrases as a unit.

A limitation of the study concerns the false-positive and false-negative indication of knowledge transfer. The diffusion of two-term combinations does not necessarily mean that knowledge transfer has taken place, as knowledge can be individually acquired without ever interacting. False-positive cases may also result from the validation of findings. The research stays in this study are almost a decade ago, so scientists may have difficulties remembering every detail of their research work abroad and rather report what comes to mind first.

Even though the bibliometric method identified internationally mobile scientists without traceable knowledge transfer, these scientists mostly replied that they acquired knowledge abroad. Several reasons were provided why no knowledge transfer was detected among this control group. In case these respondents have indeed acted as knowledge transmitters between the research group abroad and that in Germany, the method would produce false negatives. However, as there is no other method to detect knowledge transfer, it is out of scope to quantify the rate of false negatives.

Undoubtedly, internationally mobile scientists not only transfer knowledge between source and target, but also shape and modify knowledge constructs while transferring it. Enhanced bibliometric methods could provide evidence on how knowledge is altered and embedded in new contexts after successful transfer. Some knowledge-transfer processes can lead to more transformations than others. Analyses of larger samples may unveil patterns and regularities of knowledge transfer processes. Further research could also focus on the conditions under which knowledge is transferred. One assumption is that the closer the research area of source and recipient the more unhampered the knowledge transfer.

6 | CONCLUSION

Although international mobility experience is associated with positive outcomes and is a prerequisite for higher positions in the careers of scientists, we still know little about the role of international mobility in terms of

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knowledge acquisition and transfer. One major driver of the study was to develop a bibliometric method that can uncover to what extent scientists acquire knowledge abroad and pass it on. The aim of this proof-of-principle study was to demonstrate that even though bibliometric data may have limitations, they afford the detection of knowledge transfer. Methodologically, this study proposed a framework to examine knowledge flows by adopting a network perspective. The combination of rare lexical terms in abstracts of publications were used to operationalize knowledge. The major advantage of a bibliometric method that is capable of tracing knowledge transfer is that scientists do not have to be questioned directly about their knowledge acquisition. Although bibliometric studies are rarely validated with external data, this study juxtaposed bibliometric data with email responses and thus validated the developed method in an elaborate and successful way.

The results provide evidence of German scientists acting as knowledge transmitters. This issue is novel and important as there are no comprehensive methods to detect knowledge transfer. The strength of the bibliometric method is its simplicity in focusing only on lexical terms and the exclusion of noise by adjusting the TF-IDF. The validation shows that the overall precision of the two-term combinations reached 98.6%. This high precision is achieved at the expense of recall. The combination of two rare terms may reduce the fuzziness but also the probability to be used by different actors in the network. Considering the complexity of modeling knowledge transfer processes, the proposed method can indicate the transfer of tacit knowledge at least to some degree.

To conclude, the transfer of knowledge has not yet been studied comprehensively enough as the number of multivariate studies is low. Some ways forward could be the improved operationalization of knowledge contents and larger sample sizes. The findings in this study and the exploratory procedure can at best indicate individual cases but not allow for generalization. Nonetheless, the study can be seen as a step forward in identifying knowledge transmitters and revealing their specific knowledge transferred. The examples have shown that even though two-term combinations represent knowledge transfer only by approximation, they mostly matched the email replies. However, the method is not without limitations and offers plenty of room for improvement. As international mobility facilitates the transfer of knowledge and contributes to the advancement of science, it remains crucial to develop more advanced methods to understand the process of knowledge transfer across borders.

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ENDNOTE

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